hw02

Please make sure you're following all the instructions described in the "Homework Instructions" page on Canvas (go to the homepage of our class and you'll see a link saying "Homework Instructions" just underneath the header of the page).

1 Vector basics

1.1 *Vector creation*

Create a vector containing the names Frodo, Sam, Legolas, Gimli, Boromir, Pippin, Merry, Aragron, and Gandalf (in that order). Then use an appropriate function to count the number of names in the vector.

vars:

- `fellowship` - vector containing names
- `n_fellowship` - count of the number of names

1.2 *Indexing*

Use vector indexing to create a new vector containing the names Frodo, Sam, Pippin, and Merry (in that order).

var: `hobbits`

1.3 *Removing elements*

Use vector indexing (using negative indices) to create a new vector (from `fellowship`) with the names Gandalf the Grey and Boromir removed.

var: `fellowship2`

1.4 *Adding elements*

Use the `c()` function to create a new vector that's the result of appending `fellowship2` with the string "Gandalf the White".

var: `fellowship3`

2 Repetition

2.1 *Lots of fives*

Create a vector that contains the number 5 repeated 100 times (note: you shouldn't type out 100 5's - use an appropriate function).

var: `fives`

2.2 *Repeating two numbers*

Create a vector that contains 100 1's followed by 100 5's. Use a single call to `rep()`. You may want to check the documentation (`?rep`) if you're not sure how to do this. I'd recommend scrolling to the bottom to look at the examples. Beneath the "Examples" header is a "Run examples" link - if you click on this, you'll see the output from the example code. This will help you identify which options you should be using.

vars: `onefive`

3 Recycling

Create a vector containing the numbers 1 through 30 (how can we do this without typing out all 30 numbers?). Next, add 1 to every third number (elements, 3, 6, 9, etc. should each be incremented by 1). Be sure to make use of recycling.

var: `nums` - result of adding 1 to every third number of 1 through 30

4 Sequences

For this question, you'll use the `seq()` function to generate sequences of numbers. You can provide arguments to this function in a couple different ways. For example, the following are all valid usages of `seq()`:

```
seq(from = 1, to = 7, by = 3)
seq(from = 1, to = 7, length.out = 10)
seq(from = 1, by = 2, length.out = 50)
```

4.1 *Sequence 1*

Create a sequence of 23 evenly-spaced numbers where the minimum value is 1 and the maximum value is 97.

var: `seq1`

4.2 *Sequence 2*

Create a sequence of multiples of 7, starting with 7 and ending with 77.

var: `seq2`

4.3 *Sequence 3*

Create a sequence containing the first 25 multiples of 3 (it should start with 3).

var: `seq3`

5 Square root and `NaN`

5.1 *Square root*

First, use `seq()` to create a vector of all even numbers from -8 to 24 (inclusive - first number should be -8 and last number should be 24).

Then use `sqrt()` to find the square root of these numbers.

var: `sqrt_even`

5.2 *Mean*

Take the mean of the vector from Part 1. You'll need to remove the `NaN`s in order to calculate the mean - check the documentation of `mean()` (i.e. run `?mean`) to determine how to do this.

var: `sqrt_even_mean`

6 Summary statistics

Use the appropriate R functions to find the mean, standard deviation, and variance of the numbers (3,1,5,7,4,2,4,7,9).

vars:

- `mn`: the mean
- `st`: the standard deviation
- `vr`: the variance

7 Temperature conversion

(From *The Book of R* by Davies)

The conversion from a temperature measurement in degrees Fahrenheit F to Celsius C is performed using the following equation:

$$C = \frac{5}{9}(F - 32)$$

User vectorized operations in R to convert the temperatures 45, 77, 10, 19, 101, 120, and 212 in degrees Fahrenheit to degrees Celsius.

vars: `temp_c` - the temperature in Celsius

8 Miles per gallon

Suppose that for the last three times I've filled up my car with gas, I've kept track of the amount of gas I've put in my car and the number of miles I travelled on each tank of gas. Calculate the miles per gallon for each tank.

```
miles:   | 320.1 | 132.1 | 247.5 |
gallons: |   9.5 |   4.3 |   7.8 |
```

Hint: First you'll want to create a vector for miles and a vector for gallons and then use those to calculate the MPG.

vars: `mpg` - numeric vectors with 3 numbers (the MPG for each tank)